

Assignments + Python Notebook + Short Report

Week-1-Assignment

WiDS

December 2025

1 Introduction

This is the mandatory assignment related to contents of Week 1. We have tried to keep it a comprehensive assignment. Hope you enjoy solving it!

2 Check List

Use this as your check list for the week 1 contents-

- ✓ 1. Watch the IBM video attached at the end of "Introduction to AI, Machine Learning & Data Science" article.
- ✓ 2. Watch atleast the first 4 videos of playlist "AI for Everyone" by Andrew NG attached at the end of the same article.
- ✓ 3. Review Linear Algebra
- ✓ 4. Review Probability
- ✓ 5. Linear Algebra and Probability sections from the Mandatory assignment
- ✓ 6. Chapter 2 from ISLP book **⊛ 2.3 : R intro**
- ✓ 7. Linear Regression (You can read chapter 3 from ISLP, or use any other resource listed or anything else you find online)
- ✓ 8. Linear Regression section from mandatory assignment
9. Kaggle Challenge "Linearly Sparse"

3 Linear Algebra

Here are some easy Linear Algebra questions-

3.1 Solving a Linear System in an ML Context

In a simple linear regression model with two features, the normal equation can be written as

$$Aw = b$$

where

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

1. Solve for the weight vector w using the matrix inverse.
2. Write the solution explicitly in the form

$$w = A^{-1}b.$$

3. Verify your solution by substituting w back into the equation Aw .

3.2 From Scalar Derivatives to Matrix Derivatives

This question is designed to gradually build intuition from basic calculus to matrix derivatives, which are heavily used in machine learning.

Part A: Single-variable derivative

Let

$$f(x) = x^2 + 3x + 1.$$

1. Compute $\frac{df}{dx}$. $\frac{dy}{dx} = 2x + 3$

Part B: Multivariable function and partial derivatives

Now consider the function

$$g(x_1, x_2) = x_1^2 + 2x_1x_2 + 3x_2^2. \quad \frac{\partial g}{\partial x_1} = 2x_1 + 2x_2 = 2(x_1 + x_2)$$

2. Compute the partial derivatives $\frac{\partial g}{\partial x_1}$ and $\frac{\partial g}{\partial x_2}$. $\frac{\partial g}{\partial x_2} = 2x_1 + 6x_2 = 2(x_1 + 3x_2)$

Part C: Derivative of a vector-valued function

Define the vector-valued function

$$h(x) = \begin{bmatrix} x^2 \\ 3x + 1 \end{bmatrix}. \quad \frac{dh}{dx} = \begin{bmatrix} 2x \\ 3 \end{bmatrix}$$

3. Compute $\frac{dh}{dx}$.

(Hint: Differentiate each component with respect to x .)

Part D: Gradient as stacked partial derivatives

Let

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

3.1

$$Aw = b$$

$$1.) \quad A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

$$\textcircled{1} \quad A^{-1} = \frac{\text{adj}(A)}{|A|} \quad |A| = 11 \quad \text{adj}(A) = \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{3}{11} & \frac{-1}{11} \\ \frac{-1}{11} & \frac{4}{11} \end{bmatrix}$$

$$\begin{aligned} \textcircled{2} \quad w &= A^{-1}b = \frac{1}{11} \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \frac{1}{11} \begin{bmatrix} 3-2 \\ -1+8 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 1 \\ 7 \end{bmatrix} \end{aligned}$$

$$w = \begin{bmatrix} 1/11 \\ 7/11 \end{bmatrix}$$

Verifying: $Aw = b$

$$\Rightarrow \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1/11 \\ 7/11 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{4}{11} + \frac{7}{11} \\ \frac{1}{11} + \frac{21}{11} \end{bmatrix} = \begin{bmatrix} \frac{11}{11} \\ \frac{22}{11} \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad g(x_1, x_2) = x_1^2 + 2x_1x_2 + 3x_2^2$$

4. Write the gradient of $g(x_1, x_2)$ with respect to the vector x , i.e.,

$$\nabla_x g = \frac{\partial g}{\partial x} = \begin{bmatrix} \frac{\partial g}{\partial x_1} \\ \frac{\partial g}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 + 2x_2 \\ 2x_1 + 6x_2 \end{bmatrix} = 2 \begin{bmatrix} x_1 + x_2 \\ x_1 + 3x_2 \end{bmatrix}$$

(Your answer should be a 2×1 vector.)

Part E: Writing the scalar function as a matrix expression

5. Show that the function $g(x_1, x_2)$ can be written in the form $x^\top = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$

$$g(x) = x^\top A x$$

for a suitable symmetric matrix A . Explicitly find A . $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Part F: Matrix derivative

6. Using your result from Part E, compute

$$\frac{\partial}{\partial x} (x^\top A x).$$

(Hint: Your answer should be a vector of the same shape as x .)

Part G: Basic matrix derivative identities

7. Prove the following matrix derivative identities for a constant matrix A :

- (a) $\frac{\partial}{\partial x} (a^\top x) = a$
- (b) $\frac{\partial}{\partial x} (x^\top x) = 2x$
- (c) $\frac{\partial}{\partial x} (x^\top A x) = (A + A^\top)x$

4 Probability

4.1 Bayes' Rule and Classification

In a medical testing scenario, consider the following probabilities:

$$P(\text{disease}) = 0.01, \quad P(\text{positive} \mid \text{disease}) = 0.99, \quad P(\text{positive} \mid \text{no disease}) = 0.05.$$

1. Write an expression for the probability that a person has the disease given that the test result is positive, i.e.,

$$P(\text{disease} \mid \text{positive}).$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Using Baye's & Total Prob. Theorem:

Given:

$$P(d) = 0.01$$

$$P(p/d) = \frac{P(p \cap d)}{p(d)} = 0.99$$

$$\therefore p(p \cap d) = 0.99 \times 0.01$$

$$P(p/d') = 0.05$$

TPT:

$$\begin{aligned} P(p) &= P(p/d)P(d) + P(p/d')P(d') \\ &= 0.99 \times 0.01 + 0.05 \times 0.99 \end{aligned}$$

$$\begin{aligned} P(d/p) &= \frac{P(d \cap p)}{P(p)} \\ &= \frac{\cancel{0.99} \times 0.01}{\cancel{0.99} \times 0.01 + 0.05 \times \cancel{0.99}} \end{aligned}$$

$$P(d/p) = \frac{1}{6} = 0.167$$

3.2.j Part E:

$$x^T = [x_1 \ x_2] \quad \text{Let } A \text{ be a symmetric } 2 \times 2 \text{ matrix } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{Given: } g(x) = x^T A x$$

$$\therefore x_1^2 + 2x_1x_2 + 3x_2^2 = [x_1 \ x_2] \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{aligned} \text{Using Associative property: } g(x) &= x^T (A \cdot x) \\ &= [x_1 \ x_2] \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix} \\ &= [ax_1^2 + bx_1x_2 + cx_1x_2 + dx_2^2] \\ x_1^2 + 2x_1x_2 + 3x_2^2 &= [ax_1^2 + (b+c)x_1x_2 + dx_2^2] \end{aligned}$$

$$\therefore a = 1 \quad \text{Symmetric matrix } \therefore b = c \quad [i_{12} = i_{21}] \quad \therefore b = c = 1$$
$$d = 3$$

$$\therefore A = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$$

3.2} Part F:

$$\frac{\partial}{\partial x} (x^T A x) = \frac{\partial g}{\partial x} = 2 \begin{bmatrix} x_1 + x_2 \\ x_1 + 3x_2 \end{bmatrix} = 2 A x$$

$$A x = \begin{bmatrix} a x_1 + b x_2 \\ c x_1 + d x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_1 + 3x_2 \end{bmatrix}$$

Part G:

7.4

a.4 LHS: Assume 'a' is a 2×1 matrix $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ RHS: a

$$\frac{\partial}{\partial x} (a^T x) = a^T x = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$f(x_1, x_2) = \begin{bmatrix} a_1 x_1 \\ a_2 x_2 \end{bmatrix}$$

$$\therefore \frac{\partial}{\partial x} (a^T x) = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = a$$

$\therefore \text{LHS} = \text{RHS}$

Hence, proved

$$\text{b.y } \frac{\partial}{\partial x}(x^T x) = 2x$$

$$\begin{aligned} \text{LHS:} \quad & \begin{matrix} 1 \times 2 & 2 \times 1 \\ x^T x = [x_1 \ x_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ & = [x_1^2 + x_2^2] \end{matrix} \end{aligned}$$

$$\text{RHS: } 2x$$

$$\therefore \frac{\partial}{\partial x}(x^T x) = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix} = 2 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2x$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence, proved

$$\text{c.y } \frac{\partial}{\partial x}(x^T A x); \text{ for a matrix } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow x^T A x = [ax_1^2 + bx_1x_2 + cx_1x_2 + dx_2^2]$$

LHS:

$$\text{We know } Ax = \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix} \quad \& \quad A^T x = \begin{bmatrix} ax_1 + cx_2 \\ bx_1 + dx_2 \end{bmatrix}$$

$$\frac{\partial}{\partial x}(x^T A x) : \begin{bmatrix} 2ax_1 + (b+c)x_2 \\ (b+c)x_1 + 2dx_2 \end{bmatrix}$$

RHS:

$$\therefore (A + A^T)x = \begin{bmatrix} 2ax_1 + (b+c)x_2 \\ (b+c)x_1 + 2dx_2 \end{bmatrix}$$

$\therefore \text{LHS} = \text{RHS}$ Hence, proved



4.2 Maximum Likelihood Estimation for a Gaussian Model

Suppose that

$$X_1, X_2, \dots, X_n$$

are independent and identically distributed random variables such that

$$X_i \sim \mathcal{N}(\mu, \sigma^2), \quad \rightarrow \text{X follows Normal (Gaussian) distribution with mean } \mu \text{ \& var. } \sigma^2$$

where both the mean μ and the variance σ^2 are unknown.

In this question, you will derive the maximum likelihood estimators (MLEs) of μ and σ^2 .

The principle of maximum likelihood estimation is to choose the parameter values that maximize the probability of observing the given data.

1. Write the joint likelihood function

$$L(\mu, \sigma^2) = p(X_1, X_2, \dots, X_n \mid \mu, \sigma^2)$$

using the assumption that the samples are i.i.d.

2. Take the natural logarithm of the likelihood function and obtain the log-likelihood function

$$\ell(\mu, \sigma^2).$$

3. Compute the partial derivative of the log-likelihood with respect to μ , and find the value of μ that maximizes the log-likelihood.
4. Compute the partial derivative of the log-likelihood with respect to σ^2 , and find the value of σ^2 that maximizes the log-likelihood.
5. Hence, derive explicit expressions for the maximum likelihood estimators

$$\hat{\mu}_{\text{MLE}} \quad \text{and} \quad \hat{\sigma}_{\text{MLE}}^2$$

as functions of $X_1, X_2, \dots, X_n$.

didn't understand how to approach fully

attempted learning from online source

5 A Comprehensive Introduction to Linear Regression

Part : Prediction from limited observations

Suppose there is an unknown system that takes a real number as input and produces a real number as output. You are given the following input-output observations:

$$(2, 4), \quad (4, 8), \quad (9, 18)$$

→ Ans. Since we have a quantitative output we prefer to have a regression problem, we try to express the output $Y = f(X) + c$ where X is our input.

Here, since for all 3 given data points a relation $Y = 2X$ is the simplest relation that can be established, we tend to extrapolate the relation for $x = 5$.

1. Based on the above observations, what would you predict the output to be when the input is $x = 5$? → 10

Most people predict the output to be 10, i.e., twice the input.

2. Explain, in words, why you were inclined to make this prediction.

Now consider the cubic function

$$f(x) = ax^3 + bx^2 + cx + d,$$

where the coefficients a, b, c, d are chosen such that

$$f(2) = 4, \quad f(4) = 8, \quad f(9) = 18.$$

3. Does this cubic function necessarily satisfy $f(5) = 10$? No.
4. What assumption led you to believe that the output should be twice the input? That a basic linear relationship is more intuitively possible for a given data set.

This illustrates a key idea in machine learning: every model makes assumptions (hypotheses) about the underlying data-generating process.

Part: The hypothesis of linear regression

In linear regression, we assume that the output is a linear function of the input.

5. State the hypothesis made by linear regression for a single input variable x . → approx. a linear relationship b/w i/p 'x' & o/p 'Y' : $Y \approx \beta_0 + \beta_1 X$
6. Extend this hypothesis to the case where each data point has d features.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_d X_d$$

The goal of linear regression is to determine the unknown parameters (weights) of this model using the available data.

Part: Finding the optimal weights (manual derivation)

Consider a single-feature linear regression model:

$$\hat{y} = wx + b.$$

we use method of least squares i.e. square of diff. of actual o/p & predicted o/p

9. Write the total loss over the given data points using the squared error loss.

10. Find the value of w and b that minimizes this loss.

$$\underline{w = 2} \quad \underline{b = 0}$$

∴ If loss func. is defined:

$$L(w, b) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

∴ for given data pts. :

$$L(w, b) = (4 - (2w + b))^2 + (8 - (4w + b))^2 + (18 - (9w + b))^2$$

To minimize loss: $\frac{\partial L}{\partial w} = 0$ & $\frac{\partial L}{\partial b} = 0$

$$L(w, b) = (4 - (2w + b))^2 + (8 - (4w + b))^2 + (18 - (9w + b))^2$$

$$\frac{\partial L}{\partial w} = 2(4 - 2w - b)(-2) + 2(8 - 4w - b)(-4) + 2(18 - 9w - b)(-9)$$

$$= -4(4 - 2w - b) - 8(8 - 4w - b) - 18(18 - 9w - b)$$

$$= -16 + 8w + 4b - 64 + 32w + 8b - 18^2 + 162w + 18b$$

$$0 = 202w + 30b - 404$$

$$\therefore 202w + 30b = 404$$

$$\therefore 202w + 30(10 - 5w) = 404$$

$$202w + 300 - 150w = 404$$

$$52w = 104$$

$$\therefore \underline{\underline{w = 2}}$$

$$\frac{\partial L}{\partial b} = -2(4 - 2w - b) - 2(8 - 4w - b) - 2(18 - 9w - b)$$

$$= -8 + 4w + 2b - 16 + 8w + 2b - 36 + 18w + 2b$$

$$0 = 6b + 30w - 60$$

$$\underline{\underline{b = 0}}$$

$$b + 5w = 10$$

$$\therefore b = 5(2 - w)$$

Part : Matrix formulation of linear regression

We now formulate linear regression using matrix notation.

Let

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1d} \\ 1 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix} \in R^{n \times (d+1)}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

11. Write the prediction vector $\hat{y}$ in terms of X and the weight vector w (note that this is a column vector of the weights and the bias term (the constant) can be included in w itself because we have already stacked a column of 1 in X).

$$\hat{y} = w \hat{X}$$

12. Write the squared error loss using vector and matrix notation (Hint: it is the normal of a difference).

$$L(w) = \|y - \hat{y}\|^2 = \|y - w \hat{X}\|^2$$

13. Derive the normal equations and obtain the closed-form solution for w (Hint: $\|x\|^2$ can be written as $x^T x$ for a column vector x . Use the derivative properties derived in this assignment).

$$\therefore L(w) = (y - wX)^T (y - wX)$$

Now that we are at the end of this week. Ask your self these questions- what might training mean in the context of Linear Regression. What might testing mean. Yes training means "learning" those weight using the training data. It is the value of those weights you find out by optimization. And testing means finding out the error between the actual output and your predicted output.

$$L(w) = y^T y - y^T w X - (w X)^T y + (w X)^T w X$$